

# Structural Causal Graph Units for Disease-Modifying vs. Symptomatic Drug Classification

Full Mathematical Framework, Derivations,  
Redesigned Training Objective, and One-Month MVP

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# 1 Problem Statement and Motivation

The clinical distinction Disease-Modifying (DM) versus Symptomatic (SYM) is the most therapeutically critical in pharmacology, yet no computational method classifies it from graph topology and causal reasoning alone.

**Gap.** Chanda et al. (IEEE TCBB 2026) use recurrent gene frequency on shortest metapaths—correlation, not causation, and constrained to hand-crafted schemas. Their approach does not ask whether any pathway *causes* the disease; it asks only which genes appear most often.

**This work.** We define the *Structural Causal Graph Unit* (SCGU) as the minimal causal unit in a heterogeneous biomedical graph and extend Pearl’s *do*-calculus to operate on (node, edge, association) triplets. Free traversal, guided by a learned association function, discovers causal paths without any metapath schema. The central question becomes:

$$P\left(\text{DM} \mid \text{do}(v_i = v_i^*, e_i = e_i^*, \mathcal{A}_\theta(e_i, v_i) = a_i^*)\right)$$

## 2 Formal Graph Definition

**Definition 2.1** (Heterogeneous Biomedical Knowledge Graph).

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_V, \mathcal{T}_E, \phi, \psi)$$

where  $\mathcal{V}$  is the node set;  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$  is the directed edge set;  $\mathcal{T}_V$  and  $\mathcal{T}_E$  are the node-type and edge-type sets (Table 1);  $\phi : \mathcal{V} \rightarrow \mathcal{T}_V$  maps each node to its type; and  $\psi : \mathcal{E} \rightarrow \mathcal{T}_E$  maps each edge to its type.

| Symbol                                                 | Type     | Meaning                                   |
|--------------------------------------------------------|----------|-------------------------------------------|
| $\mathcal{V}$                                          | set      | all biological entities (nodes)           |
| $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ | set      | directed biological relations (edges)     |
| $\mathcal{T}_V$                                        | set      | {Drug, Gene, Pathway, Disease, Biomarker} |
| $\mathcal{T}_E$                                        | set      | 18 typed relations (Table 1)              |
| $\phi : \mathcal{V} \rightarrow \mathcal{T}_V$         | function | maps every node to its biological type    |
| $\psi : \mathcal{E} \rightarrow \mathcal{T}_E$         | function | maps every edge to its relation type      |
| $\mathbf{x}_v \in \mathbb{R}^{d_{\phi(v)}}$            | vector   | type-specific feature vector of node $v$  |
| $c_e \in [0, 1]$                                       | scalar   | database confidence score of edge $e$     |

### Why This Is Required

A homogeneous graph collapses all biological relationships into a single edge type. The relationship *Drug inhibits Gene* and *Gene regulates Pathway* carry entirely different biological meanings. Encoding type through  $\phi$  and  $\psi$  is what allows the association function  $\mathcal{A}_\theta$  (Section 4) to distinguish (causal\_of, Pathway, Disease) from

(causal\_of, Disease, Pathway) and assign them very different causal weights.

Table 1: Node types  $\mathcal{T}_V$ , edge types  $\mathcal{T}_E$ , and primary data sources

| Source type    | Relation             | Target type    | Source database                |
|----------------|----------------------|----------------|--------------------------------|
| Drug           | binds                | Gene           | ChEMBL 34                      |
| Drug           | inhibits             | Gene           | ChEMBL 34                      |
| Drug           | activates            | Gene           | ChEMBL 34                      |
| Drug           | upregulates          | Gene           | DrugBank 5.1+                  |
| Drug           | downregulates        | Gene           | DrugBank 5.1+                  |
| Drug           | modulates            | Pathway        | DrugBank 5.1+                  |
| Drug           | treats               | Disease        | DrugBank / FDA labels          |
| Gene           | involved_in          | Pathway        | Reactome v88                   |
| Gene           | regulates            | Pathway        | Reactome v88                   |
| Gene           | catalyzes            | Pathway        | Reactome v88                   |
| Gene           | ppi                  | Gene           | STRING v12                     |
| Gene           | coexpression         | Gene           | GTEX v8                        |
| Gene           | linked_to            | Disease        | DisGeNET v7+                   |
| Gene           | biomarker_of         | Disease        | Open Targets 24.06             |
| Pathway        | associated_with      | Disease        | KEGG DISEASE                   |
| <b>Pathway</b> | <b>causal_of</b>     | <b>Disease</b> | <b>KEGG DISEASE + PharmGKB</b> |
| Pathway        | affects_trajectory   | Biomarker      | ADNI / PPMI                    |
| Biomarker      | predicts_progression | Disease        | ADNI / PPMI                    |

## 2.1 Node Feature Vectors

Each node  $v \in \mathcal{V}$  carries a type-specific feature vector  $\mathbf{x}_v \in \mathbb{R}^{d_{\phi(v)}}$ :

$$\mathbf{x}_v^{\text{Drug}} = \left[ \underbrace{\text{ECFP4}(v)}_{2048} \parallel \underbrace{\text{MolBERT}(v)}_{768} \parallel \underbrace{\text{ATC}(v)}_{87} \right] \in \mathbb{R}^{2903} \quad (1)$$

$$\mathbf{x}_v^{\text{Gene}} = \left[ \underbrace{\text{ESM-2}(v)}_{1280} \parallel \underbrace{\text{GTEX}(v)}_{54} \parallel \underbrace{\text{GO}(v)}_{1000} \right] \in \mathbb{R}^{2334} \quad (2)$$

$$\mathbf{x}_v^{\text{Pathway}} = \left[ \underbrace{\text{Membership}(v)}_{20000} \parallel \underbrace{\text{P2Vec}(v)}_{128} \right] \in \mathbb{R}^{20128} \quad (3)$$

$$\mathbf{x}_v^{\text{Disease}} = \left[ \underbrace{\text{DO2Vec}(v)}_{256} \parallel \underbrace{\text{ICD}(v)}_{256} \parallel \underbrace{\text{BioProfile}(v)}_{d_b} \right] \quad (4)$$

$$\mathbf{x}_v^{\text{Biomarker}} = \text{Trajectory}(v) \in \mathbb{R}^{T \times d_m} \quad (5)$$

where  $\text{GO}(v)$  is a binary vector over the 1000 most frequent GO biological process terms;  $\text{GTEX}(v)$  contains median TPM across 54 tissues; and  $\text{Trajectory}(v)$  is a time-series of longitudinal biomarker measurements from ADNI (Alzheimer’s) or PPMI (Parkinson’s).

**In Plain English**

Think of these feature vectors as identity cards for each biological entity. A drug like Lecanemab gets a 2903-dimensional card: ECFP4 (2048 numbers) encodes its molecular shape—which protein pockets it fits into; MolBERT (768 numbers) captures chemical language learned from millions of molecules; ATC (87 bits) says it belongs to the anti-Alzheimer’s class. A gene like BACE1 gets a 2334-dimensional card: ESM-2 (1280 numbers) encodes the 3D shape of the protein it produces; GTEx (54 numbers) records how strongly it is expressed in each of 54 body tissues— BACE1 is highest in the brain; GO (1000 bits) marks which biological jobs it does (here: aspartyl protease, amyloid precursor protein cleavage). The model needs these rich identities because two nodes in identical graph positions can have completely different biological roles—topology alone is not enough.

## 2.2 Edge Property Tuple

Each edge  $e = (v_s, v_t) \in \mathcal{E}$  carries:  $\mathbf{e} = (\psi(e), c_e, \delta_e, \epsilon_e)$  where  $c_e \in [0, 1]$  is the database confidence score;  $\delta_e \in \{+1, -1\}$  is directionality; and  $\epsilon_e \in \{\text{experimental}, \text{literature}, \text{computational}\}$  is evidence type. The confidence  $c_e$  seeds the association function initialization (Equation 12).

## 3 Structural Causal Graph Unit (SCGU)

### 3.1 Definition and Symbol Table

**Definition 3.1** (SCGU). *For a directed edge  $e_i = (v_{i-1}, v_i) \in \mathcal{E}$ , the SCGU at step  $i$  is:*

$$\tau_i = (v_i, e_i, a_i) \quad (6)$$

| Symbol                        | Space         | Meaning                            | Example                                         |
|-------------------------------|---------------|------------------------------------|-------------------------------------------------|
| $v_i$                         | $\mathcal{V}$ | target node at step $i$            | BACE1 gene                                      |
| $v_{i-1}$                     | $\mathcal{V}$ | source node at step $i$            | Lecanemab (drug)                                |
| $e_i = (v_{i-1}, v_i)$        | $\mathcal{E}$ | directed edge                      | Lecanemab $\xrightarrow{\text{inhibits}}$ BACE1 |
| $\phi(v_i) \in \mathcal{T}_V$ | type          | biological type of $v_i$           | <b>Gene</b>                                     |
| $\psi(e_i) \in \mathcal{T}_E$ | type          | relation type of $e_i$             | <b>inhibits</b>                                 |
| $a_i \in [0, 1]$              | scalar        | typed association score; see Eq. 7 | 0.91                                            |

**In Plain English**

Standard causal models treat biology like a dots-and-lines map: only the dots (nodes) matter. The SCGU says the lines matter just as much. If you are investigating whether a drug causes disease modification, you need to know not just *which* gene it connects to, but *how* it connects. Does it **inhibit** the gene (shutting it down)

or **activate** it (turning it up)? Those lead to completely different biological consequences. The triplet  $\tau_i = (v_i, e_i, a_i)$  captures: the destination node, the connection type, and a score for how much biological sense that connection makes. Lecanemab  $\xrightarrow{\text{inhibits}}$  BACE1 reduces amyloid production. Lecanemab  $\xrightarrow{\text{activates}}$  BACE1 would increase it. Same nodes, opposite biology—the SCGU distinguishes them; standard nodes-only models cannot.

### Novel Contribution

Pearl’s SCM makes *nodes* the causal primitive; edges are passive. The SCGU promotes the edge type  $\psi(e_i)$  and its semantic coupling  $a_i$  to *first-class causal variables*. Whether a drug **inhibits** vs. **activates** a target is as causally decisive as which target it acts on. An intervention on  $\tau_i$  simultaneously fixes the node value, the edge type, and their typed association—capturing the biological reality that *how* a drug connects to a target is as important as *which* target it connects to. No existing method (DoWhy, CausalNex, HAN, MAGNN) supports interventions on edge types.

## 4 Association Function $\mathcal{A}_\theta$

### 4.1 Formula

**Definition 4.1** (Typed Association Function). *The association function is a learned mapping from a typed (edge, source-node, target-node) triplet to a semantic compatibility score:  $\mathcal{A}_\theta : \mathcal{T}_E \times \mathcal{T}_V \times \mathcal{T}_V \rightarrow [0, 1]$ , parameterized as:*

$$\mathcal{A}_\theta(t_e, t_s, t_t) = \sigma\left(W_{\mathcal{A}} \cdot [\mathbf{e}_{t_e} \parallel \mathbf{e}_{t_s} \parallel \mathbf{e}_{t_t}] + b_{\mathcal{A}}\right) \quad (7)$$

| Symbol                                           | Dimension            | Meaning               | Learned? |
|--------------------------------------------------|----------------------|-----------------------|----------|
| $t_e = \psi(e_i)$                                | $\in \mathcal{T}_E$  | edge relation type    | —        |
| $t_s = \phi(v_{i-1})$                            | $\in \mathcal{T}_V$  | source node type      | —        |
| $t_t = \phi(v_i)$                                | $\in \mathcal{T}_V$  | target node type      | —        |
| $\mathbf{e}_t \in \mathbb{R}^{d_t}$              | $d_t = 64$           | type embedding vector | yes      |
| $W_{\mathcal{A}} \in \mathbb{R}^{1 \times 3d_t}$ | 192 weights          | linear projection     | yes      |
| $b_{\mathcal{A}} \in \mathbb{R}$                 | scalar               | bias                  | yes      |
| $\sigma(z) = 1/(1 + e^{-z})$                     | $\rightarrow (0, 1)$ | sigmoid activation    | —        |

**Interpretation.**

$\mathcal{A}_\theta(\text{causal\_of}, \text{Pathway}, \text{Disease}) \approx 0.93$  (biologically valid, strong DM signal)

$\mathcal{A}_\theta(\text{inhibits}, \text{Disease}, \text{Drug}) \approx 0.02$  (invalid direction, pruned)

$\mathcal{A}_\theta(\text{causal\_of}, \text{Drug}, \text{Drug}) \approx 0.01$  (biologically meaningless, pruned)

**Derivation**

**Step 1: concatenation.** Let  $\mathbf{f} = [\mathbf{e}_{t_e} \parallel \mathbf{e}_{t_s} \parallel \mathbf{e}_{t_t}] \in \mathbb{R}^{3d_t}$ . Each column block of  $W_{\mathcal{A}}$  ( $d_t$  columns each) learns how much the edge type, source type, and target type independently contribute to compatibility. Cross-type interactions are encoded by the joint gradient update.

**Step 2: linear projection.**  $s = W_{\mathcal{A}}\mathbf{f} + b_{\mathcal{A}} \in \mathbb{R}$  is a scalar compatibility score.

**Step 3: sigmoid.**  $\sigma(s) \in (0, 1)$  maps the raw score to a probability-like gate. Sigmoid is chosen over softmax (cannot gate individual steps independently) and over ReLU (unbounded, cannot serve as a semantic gate in  $[0, 1]$ ).

**Result:**  $\mathcal{A}_{\theta} \approx 1$  when the relation type  $t_e$  is a biologically valid connector from type  $t_s$  to type  $t_t$ ;  $\mathcal{A}_{\theta} \approx 0$  otherwise.

## 4.2 Gradients of the Association Function

Let  $\delta = \sigma(W_{\mathcal{A}}\mathbf{f} + b_{\mathcal{A}})$  and  $\delta' = \delta(1 - \delta)$ . Then:

$$\frac{\partial \mathcal{A}_{\theta}}{\partial W_{\mathcal{A}}} = \delta' \cdot \mathbf{f}^{\top} \in \mathbb{R}^{1 \times 3d_t} \quad (8)$$

$$\frac{\partial \mathcal{A}_{\theta}}{\partial \mathbf{e}_{t_e}} = \delta' \cdot W_{\mathcal{A}}[:, 0 : d_t]^{\top} \quad (9)$$

$$\frac{\partial \mathcal{A}_{\theta}}{\partial \mathbf{e}_{t_s}} = \delta' \cdot W_{\mathcal{A}}[:, d_t : 2d_t]^{\top} \quad (10)$$

$$\frac{\partial \mathcal{A}_{\theta}}{\partial \mathbf{e}_{t_t}} = \delta' \cdot W_{\mathcal{A}}[:, 2d_t : 3d_t]^{\top} \quad (11)$$

These gradients flow through every path step during backpropagation, jointly training the type embeddings and  $W_{\mathcal{A}}$  from the classification signal.

## 4.3 Initialization from Graph Frequencies

Before gradient descent, seed  $\mathcal{A}_{\theta}$  from observed graph frequencies:

$$\mathcal{A}_{\theta}^{(0)}(t_e, t_s, t_t) = \frac{|\{e \in \mathcal{E} : \psi(e) = t_e, \phi(v_s) = t_s, \phi(v_t) = t_t\}|}{|\{e \in \mathcal{E} : \phi(v_s) = t_s, \phi(v_t) = t_t\}|} \quad (12)$$

This encodes domain knowledge: if 91% of Pathway  $\rightarrow$  Disease edges in the observed KG are typed `causal_of`, then  $\mathcal{A}_{\theta}^{(0)}(\text{causal\_of}, \text{Pathway}, \text{Disease}) = 0.91$ . The model thus starts with accurate biological priors before any training example is seen.



**In Plain English**

This is the model’s biological common sense. Before looking at any specific drug, the model learns which types of connections are meaningful in biology. “Drug **inhibits** Gene” is meaningful ( $\mathcal{A}_\theta \approx 0.91$ ). “Disease **inhibits** Drug” is backwards and meaningless ( $\mathcal{A}_\theta \approx 0.02$ ). The function takes three inputs—the relationship type, the source node type, and the target node type—and outputs a single number between 0 and 1: “Does this connection make biological sense?”

It works like a grammar checker for biological pathways. Just as a grammar checker flags “The gene was inhibited by the disease toward the drug” as backwards,  $\mathcal{A}_\theta$  flags edges that traverse the graph in biologically invalid directions—and the beam search prunes any path containing such a step. The key practical gain: you never have to write a single metapath schema. The model learns all of this automatically from observed graph statistics.

**Why This Is Required**

$\mathcal{A}_\theta$  is the mechanism that replaces predefined metapath schemas. On Hetionet (47k nodes, 2.25M edges), unconstrained traversal to depth 6 produces  $\mathcal{O}(10^{15})$  candidate paths—computationally intractable. Predefined metapath schemas (the Chanda approach) reduce this but exclude unknown biological mechanisms.  $\mathcal{A}_\theta$  provides a differentiable, learnable gate: paths with low association scores accumulate near-zero path confidence  $\text{Conf}(\pi)$  and are pruned by the beam search (Algorithm 2). The result is a path space that is biologically constrained without being biologically predetermined.

## 4.4 Semantic Gating Proposition

**Proposition 4.1** (Association Replaces Metapaths). *In Algorithm 2, any path step  $(v_{i-1}, e_i, v_i)$  with  $\mathcal{A}_\theta(\psi(e_i), \phi(v_{i-1}), \phi(v_i)) < \varepsilon_c$  is pruned.*

*Proof.* By line 9 of Algorithm 2,  $\text{Conf}(\pi) \leq \varepsilon_c$  causes the path to be discarded. Since  $\text{Conf}(\pi) = (\prod_{i=1}^k a_i)^{1/k}$  and  $a_i \geq 0$ , any single  $a_i < \varepsilon_c^k \leq \varepsilon_c$  forces  $\text{Conf}(\pi) < \varepsilon_c$ . Thus invalid steps eliminate the path regardless of the remaining steps.  $\square$

## 5 Extended Do-Operator on SCGUs

### 5.1 Pearl’s Operator: Quick Review

In a SCM  $\mathcal{M} = (\mathbf{V}, \mathbf{U}, \mathcal{F}, P(\mathbf{U}))$ , the operation  $\text{do}(X = x)$  produces mutilated model  $\mathcal{M}_x$  by: (i) setting  $X \leftarrow x$ ; and (ii) severing all edges into  $X$  in the causal DAG  $\mathcal{D}$ .

**Limitation:** the operator acts only on *node* values.

## 5.2 SCGU Do-Operator

We augment  $\mathcal{D}$  to  $\mathcal{D}^+$  by treating each edge  $e_i$  as a vertex with incoming edges from type-determining context.

**Definition 5.1** (SCGU Do-Operator).  $\text{do}(\tau_i = \tau_i^*)$  where  $\tau_i^* = (v_i^*, e_i^*, a_i^*)$  applies:

- (a) **Node intervention** (Pearl, unchanged): remove all edges into  $v_i$  in  $\mathcal{D}^+$ ; set  $v_i \leftarrow v_i^*$ .
- (b) **Edge-type intervention** (novel): remove all type-determining edges into  $e_i$  in  $\mathcal{D}^+$ ; set  $\psi(e_i) \leftarrow \psi(e_i^*)$ .
- (c) **Association fixation** (deterministic from (a)–(b)): set  $a_i \leftarrow \mathcal{A}_\theta(\psi(e_i^*), \phi(v_{i-1}), \phi(v_i))$ .

| Operation                  | Acts on     | Cuts                            | Novel?     |
|----------------------------|-------------|---------------------------------|------------|
| (a) node intervention      | $v_i$       | incoming edges to $v_i$         | No (Pearl) |
| (b) edge-type intervention | $\psi(e_i)$ | type-determining edges to $e_i$ | <b>Yes</b> |
| (c) association fixation   | $a_i$       | (determined by (a)–(b))         | <b>Yes</b> |

### Novel Contribution

Operation (b) is the key departure from all existing causal graph methods. Standard methods (DoWhy, CausalNex, GNN-based causal inference) intervene only on node values. Here, fixing the edge type  $\psi(e_i^*)$  means asking: “What would the causal effect be if this relationship were of type `causal_of` rather than `associated_with`?” This is precisely the question that distinguishes DM from SYM mechanisms: a drug acting through a pathway that `causal_of` the disease ( $a_i$  high) is evidence of DM; a drug acting through a pathway that merely `associated_with` the disease ( $a_i$  moderate) is evidence of SYM.

### In Plain English

Standard statistics asks: “what happens when we *observe* a drug being taken?” The do-operator asks: “what would happen if we *forced* this drug to act through this exact mechanism—say, through a pathway that causally drives the disease rather than one that merely correlates with it?”

The difference is exactly the gap between a survey and a randomised clinical trial. In a survey, confounders distort everything. A forced intervention removes them. Our extension goes further than Pearl: we force not just the node state (which target the drug hits) but also the edge type (how it hits that target) and the association score (how biologically valid that mechanism is). This is like conducting a surgical trial inside the graph—asking what the disease outcome would be if we could precisely control the entire drug-target relationship, not just the drug dosage.

### 5.3 Backdoor Adjustment: Full Derivation

**Definition 5.2** (Typed Backdoor Criterion).  $Z_i \subseteq \mathcal{V}$  satisfies the typed backdoor criterion for  $(\tau_i, v_i)$  if:

(i) No element of  $Z_i$  is a descendant of  $v_{i-1}$  in  $\mathcal{D}^+$ .

(ii)  $Z_i$  blocks all backdoor paths from  $\tau_i$  to  $v_i$  in  $\mathcal{D}^+$ .

**Theorem 5.1** (Identifiability via Typed Backdoor Adjustment). If  $Z_i$  satisfies the typed backdoor criterion:

$$P(v_i \mid \text{do}(\tau_i)) = \sum_{z \in \mathcal{Z}_i} P(v_i \mid v_{i-1}, e_i, a_i, Z_i = z) \cdot P(Z_i = z) \quad (13)$$

#### Derivation

**Step 1.**  $\text{do}(\tau_i)$  severs all incoming edges to  $v_i$  and all type-determining edges to  $e_i$  in  $\mathcal{D}^+$ .

**Step 2.** After mutilation, the only directed paths from the intervention to  $v_i$  are those passing through  $v_{i-1} \rightarrow e_i \rightarrow v_i$ .

**Step 3.** If  $Z_i$  d-separates all backdoor paths in  $\mathcal{D}_{\tau_i}^+$ , then for any  $Z_i = z$ :

$$P_{\mathcal{D}_{\tau_i}^+}(v_i \mid v_{i-1}, e_i, a_i, Z_i = z) = P(v_i \mid v_{i-1}, e_i, a_i, Z_i = z)$$

**Step 4.** Marginalize over  $Z_i$  to obtain Equation 13. This follows from Pearl (2009), Theorem 3.3.2, applied to  $\mathcal{D}^+$ .

#### Why This Is Required

The KG is built from observational data. A gene may correlate with both drug binding *and* disease state not because the drug acts causally through it, but because a shared upstream regulator drives both. Without backdoor adjustment, the model learns spurious correlations. Equation 13 removes this bias by adjusting for confounders  $Z_i$  before computing the causal probability.

#### In Plain English

Confounders in biology are everywhere. A gene might be correlated with both a drug's target and disease progression not because the drug acts causally through it, but because a shared upstream regulator—a transcription factor active in the disease state—drives both. If we just look at correlations (what the model observes), we conclude the drug acts through that gene causally. Wrong.

The backdoor adjustment is the mathematical equivalent of holding confounders constant in a clinical trial. We identify the upstream regulators  $Z_i$ , condition on them, and *then* ask: what is the drug's effect? After conditioning, the confounding disappears. What remains is the genuine causal signal. The result is the same as what

you would get from a randomised controlled trial—but computed entirely from graph data.

## 5.4 Practical Approximation

$$P(v_i \mid \text{do}(\tau_i)) \approx \frac{1}{|\hat{Z}_i|} \sum_{z \in \hat{Z}_i} P_\theta(v_i \mid v_{i-1}, e_i, a_i, z) \quad (14)$$

where  $\hat{Z}_i$  is the 1-hop KG query set of common upstream regulators of both endpoints of  $e_i$  (typically 2–15 nodes), and  $P_\theta(v_i \mid \dots) = \sigma(\mathbf{w}_s^\top [\mathbf{h}_{v_i}^{(L)} \parallel \mathbf{h}_{v_{i-1}}^{(L)} \parallel \mathbf{h}_{e_i}^{(L)} \parallel a_i] + b_s)$ .

# 6 Causal Path and Effect

## 6.1 Path Definition

**Definition 6.1** (SCGU Path).

$$\pi = (v_0 = d, \tau_1, v_1, \tau_2, v_2, \dots, \tau_k, v_k = \text{dis}), \quad |\pi| = k$$

## 6.2 Causal Effect $\text{CE}(\pi)$ : Full Derivation

**Assumption 6.1** (SCGU Markov Property).  $P(v_i \mid \text{do}(\tau_i, \dots, \tau_1), v_{i-1}) = P(v_i \mid \text{do}(\tau_i), v_{i-1})$  for all  $i$ .

*This holds when the biological effect of one step is mediated entirely through the resulting node state, not memory of earlier steps—standard in biological signaling.*

### Derivation

Under Assumption 6.1, the joint causal probability factorizes:

$$P(v_1, \dots, v_k \mid \text{do}(\tau_1, \dots, \tau_k), v_0) = \prod_{i=1}^k P(v_i \mid \text{do}(\tau_i), v_{i-1})$$

$$\boxed{\text{CE}(\pi) = \prod_{i=1}^k P(v_i \mid \text{do}(\tau_i), v_{i-1})} \quad (15)$$

| Symbol                                   | Range             | Meaning                                    | High value means    |
|------------------------------------------|-------------------|--------------------------------------------|---------------------|
| $\text{CE}(\pi)$                         | $(0, 1]$          | product of step-level causal probabilities | strong causal chain |
| $P(v_i \mid \text{do}(\tau_i), v_{i-1})$ | $(0, 1)$          | causal probability at step $i$             | causally valid step |
| $k =  \pi $                              | $\{1, \dots, 6\}$ | path length                                | more hops           |

**In Plain English**

Think of a causal path as a chain of dominoes from the drug all the way to the disease. Each domino falls (each biological step is causally triggered) with some causal probability—computed via the backdoor-adjusted formula above. The total causal effect of the path is the *product* of all these probabilities, exactly like compound probability.

If every link has 80% causal probability across 4 steps:  $0.8^4 = 0.41$ . Only paths where every single link is genuinely causal will produce high CE scores. A path that goes Drug  $\rightarrow$  Gene (CE: 0.9)  $\rightarrow$  Pathway (CE: 0.8)  $\rightarrow$  Disease (CE: 0.7) has overall CE =  $0.9 \times 0.8 \times 0.7 = 0.50$ . One weak link (say 0.2) collapses the whole path to  $\leq 0.20$ . This is intentional: weak causal links anywhere in the mechanism are red flags.

**6.2.1 Log-Space Computation for Numerical Stability**

$$\log \text{CE}(\pi) = \sum_{i=1}^k \log P(v_i \mid \text{do}(\tau_i), v_{i-1}) \quad (16)$$

then  $\text{CE}(\pi) = \exp(\log \text{CE}(\pi))$  with clipping at  $e^{-30}$  to prevent NaN gradients.

**6.2.2 Gradient of  $\text{CE}(\pi)$  via the Log-Derivative Trick**

$$\frac{\partial \text{CE}(\pi)}{\partial \theta} = \text{CE}(\pi) \cdot \sum_{i=1}^k \frac{\partial \log P(v_i \mid \text{do}(\tau_i), v_{i-1})}{\partial \theta} \quad (17)$$

**6.3 Path Confidence  $\text{Conf}(\pi)$** 

$$\text{Conf}(\pi) = \left( \prod_{i=1}^k a_i \right)^{1/k} \quad (18)$$

**Derivation**

**Why geometric mean, not arithmetic mean?** If one step has  $a_j = 0.01$  and all others  $a_i = 0.99$ :

$$\bar{a}_{\text{arith}} \approx 0.83 \quad (k = 6)$$

$$\bar{a}_{\text{geom}} \approx 0.38$$

By AM-GM inequality,  $\bar{a}_{\text{geom}} \leq \bar{a}_{\text{arith}}$  always. The geometric mean is the correct choice because *one biologically invalid traversal step invalidates the entire proposed mechanism*, not merely lowers the average.

### In Plain English

CE measures how causally strong a path is. Conf measures how biologically plausible each step is—even before asking about causality. A path could have high CE (statistically consistent causal chain) but pass through a connection that is biologically backwards, like a disease activating a drug. Conf catches that.

The geometric mean is the right tool because biology has a zero-tolerance policy for invalid steps. One nonsensical connection in a six-step path does not just lower the average; it calls the entire proposed mechanism into question. The geometric mean reflects this: one step with  $a_j = 0.01$  drags Conf down to roughly 0.38 across six steps (see the derivation). Compare: the arithmetic mean would give 0.83—far too generous to a biologically invalid mechanism.

## 7 DM-SYM Score

### 7.1 Formula and Symbol Table

$$\text{DM}_{\text{score}}(d) = \frac{1}{Z_d} \sum_{\pi \in \Pi(d)} \underbrace{|\pi|^{-\lambda}}_{\text{(A) length}} \cdot \underbrace{\text{CE}(\pi)}_{\text{(B) causal}} \cdot \underbrace{\text{Conf}(\pi)}_{\text{(C) semantic}} \quad (19)$$

$$Z_d = \sum_{\pi \in \Pi(d)} |\pi|^{-\lambda} \cdot \text{Conf}(\pi)$$

| Term               | Value/Range                | Meaning                                                                                                      | Why needed                          |
|--------------------|----------------------------|--------------------------------------------------------------------------------------------------------------|-------------------------------------|
| $\Pi(d)$           | set of paths               | causally valid paths from drug $d$ to disease, $\text{Conf}(\pi) > \varepsilon_c$ ,<br>$ \pi  \leq L_{\max}$ | domain of summation                 |
| $ \pi ^{-\lambda}$ | $(0, 1]$ , $\lambda = 0.5$ | length penalty; $ \pi  = 6 \Rightarrow 0.41$                                                                 | offset spurious long-path inflation |
| $\text{CE}(\pi)$   | $(0, 1]$                   | product of step-level causal probabilities                                                                   | causal quality of path              |
| $\text{Conf}(\pi)$ | $(0, 1]$                   | geometric mean of association scores                                                                         | biological validity gate            |
| $Z_d$              | $> 0$                      | normalizing constant                                                                                         | removes promiscuity bias            |

### Derivation

**Probabilistic interpretation.** Treat each path  $\pi$  as a hypothesis. Define the prior  $q(\pi) \propto |\pi|^{-\lambda} \cdot \text{Conf}(\pi)$ , normalized by  $Z_d$ . Then:

$$\text{DM}_{\text{score}}(d) = \mathbb{E}_{q(\pi)}[\text{CE}(\pi)]$$

A DM drug achieves high score when its most biologically plausible paths carry high

causal effect. Without  $Z_d$ , a broad-spectrum drug targeting 200 genes accumulates a larger raw sum than a targeted drug with 3 causal paths, purely due to quantity.  $Z_d$  ensures the score reflects path *quality*, not count.

### In Plain English

This is the final verdict. After discovering all paths from a drug to its disease targets, the model aggregates all the evidence into one number.

Think of it as a weighted jury vote: each path casts a vote for “DM” weighted by (A) how direct it is—shorter paths vote more strongly; (B) how causally strong it is—high-CE paths vote more strongly; (C) how biologically plausible every step is—high-Conf paths vote more strongly. The normaliser  $Z_d$  ensures a promiscuous drug targeting 200 genes does not automatically win just because it has more paths.

**What you expect to see:**

- **DM drugs** (e.g. Lecanemab): high  $\text{DM}_{\text{score}}$ , driven by a few short paths through `causal_of` edges.
- **SYM drugs** (e.g. Donepezil): low  $\text{DM}_{\text{score}}$ , paths dominated by `associated_with` edges, no direct causal route to disease modification.

## 7.2 Gradient of DM Score

$$\frac{\partial \text{DM}_{\text{score}}(d)}{\partial \theta} = \frac{1}{Z_d} \sum_{\pi \in \Pi(d)} |\pi|^{-\lambda} \cdot \text{Conf}(\pi) \cdot \frac{\partial \text{CE}(\pi)}{\partial \theta} \quad (20)$$

## 7.3 Classification Rule

$$\hat{y}(d) = \begin{cases} \text{DM} & \text{if } \text{DM}_{\text{score}}(d) > \theta^* \\ \text{SYM} & \text{otherwise} \end{cases} \quad \text{where } \theta^* = \arg \max_{\theta} [\text{Sens}(\theta) + \text{Spec}(\theta) - 1]$$

learned on the validation set via Youden’s index.

# 8 SCGU-GNN Architecture

## 8.1 Message Passing with Causal Attention

$$\mathbf{h}_{v_i}^{(l+1)} = \sigma \left( W_{\text{self}}^l \cdot \mathbf{h}_{v_i}^{(l)} + \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{\text{causal}} \cdot \mathbf{m}_{ij}^l \right) \quad (21)$$

| Symbol                                              | Dimension        | Meaning                                 |
|-----------------------------------------------------|------------------|-----------------------------------------|
| $\mathbf{h}_{v_i}^{(l)} \in \mathbb{R}^{d_h}$       | $d_h = 256$      | node $v_i$ embedding at layer $l$       |
| $W_{\text{self}}^l \in \mathbb{R}^{d_h \times d_h}$ | $256 \times 256$ | self-loop weight matrix                 |
| $\mathcal{N}(i)$                                    | set              | in-neighbours of $v_i$ in $\mathcal{G}$ |
| $\alpha_{ij}^{\text{causal}} \in (0, 1)$            | scalar           | causal attention weight (Eq. 23)        |
| $\mathbf{m}_{ij}^l \in \mathbb{R}^{d_h}$            | 256              | SCGU message from $v_j$ (Eq. 22)        |

### In Plain English

At each layer, every node updates its understanding of itself by collecting information from its neighbours. After  $L = 3$  layers, a drug node has aggregated information from up to 3 hops away—it knows about its direct targets (1 hop), the pathways those genes are involved in (2 hops), and the diseases those pathways affect (3 hops). This is exactly the Drug  $\rightarrow$  Gene  $\rightarrow$  Pathway  $\rightarrow$  Disease chain.

The self-loop term  $W_{\text{self}}^l \mathbf{h}_{v_i}^{(l)}$  preserves the node’s own information across layers, preventing the GNN from forgetting what the node originally was as it accumulates neighbour context.

## 8.2 SCGU Message

$$\mathbf{m}_{ij}^l = W_{\text{msg}}^l \cdot [\mathbf{h}_{v_j}^{(l)} \parallel \mathbf{h}_{e_{ij}}^{(l)} \parallel a_{ij}] \quad (22)$$

where  $\mathbf{h}_{e_{ij}}^{(l)}$  is the edge-type embedding and  $a_{ij}$  is appended as a scalar. The message concatenates (1) what the source node is, (2) how it is related, and (3) how semantically valid this pairing is. Without the association scalar, the message conflates **inhibits** with **activates**.

### In Plain English

When BACE1 (gene node) receives a message from Lecanemab (drug node), the message carries three pieces of information: *who is sending* (Lecanemab’s molecular embedding), *how they are connected* (the **inhibits** edge type embedding), and *whether that connection makes sense* ( $a_{ij} = 0.91$ ). Without the edge embedding, the message conflates **inhibits** with **activates**. Without the association scalar, it cannot tell whether that particular drug-gene connection is biologically meaningful. All three travel together through  $W_{\text{msg}}^l$ .

## 8.3 Causal Attention

$$\alpha_{ij}^{\text{causal}} = \text{softmax}_j \left( \underbrace{\text{LReLU}(\mathbf{q}^\top [W_Q \mathbf{h}_{v_i}^{(l)} \parallel W_K \mathbf{h}_{v_j}^{(l)}])}_{s_{ij}: \text{feature similarity (HAN)}} + \underbrace{\beta \cdot a_{ij}}_{\text{causal bias (novel)}} \right) \quad (23)$$



| Symbol                                     | Dimension             | Meaning                         |
|--------------------------------------------|-----------------------|---------------------------------|
| $\mathbf{q} \in \mathbb{R}^{2d_h}$         | 512                   | attention query vector          |
| $W_Q, W_K \in \mathbb{R}^{d_h \times d_h}$ | $256 \times 256$ each | query/key projection            |
| LReLU                                      | —                     | LeakyReLU with slope 0.2        |
| $\beta \in \mathbb{R}$                     | scalar                | learned association bias weight |

### Novel Contribution

**Comparison with HAN and MAGNN.** HAN attention is:  $\alpha_{ij}^{\text{HAN}} = \text{softmax}_j(\text{LReLU}(\mathbf{a}^\top [W\mathbf{h}_{v_i}^{(l)} \| W\mathbf{h}_{v_j}^{(l)}]))$  and requires predefined metapath schemas to determine which neighbors to include. MAGNN uses metapath instance encoders over predefined schemas. Both methods hard-code biological assumptions. In Equation 23,  $\beta \cdot a_{ij} \rightarrow 0$  when  $a_{ij} \approx 0$  (semantically invalid pairing), suppressing the message regardless of feature similarity—*without any predefined schema*. The additive (not multiplicative) bias preserves the feature similarity signal while layering the causal constraint on top.

### In Plain English

Standard graph attention (HAN) asks: “how similar are your features to mine?” and assigns high weight to similar neighbours. The problem: two nodes can be feature-similar but connected by a biologically nonsensical edge type, and HAN would still pass that message.

Causal attention adds a second question: “is this connection biologically valid?” The term  $\beta \cdot a_{ij}$  adds an association-based bonus or penalty to the standard attention score. When  $a_{ij} \approx 0$  (invalid pairing—say, a Disease node trying to activate a Drug node), the attention weight collapses toward zero *regardless of feature similarity*. When  $a_{ij} \approx 1$  (valid pairing), the message is amplified. The learnable  $\beta$  controls how strongly biological validity overrides feature similarity. If  $\beta \rightarrow 0$ , the model becomes standard HAN and needs schemas again. If  $|\beta| \gg 1$ , the association function completely dominates.

**Gradient of causal attention.** Let  $u_{ij} = s_{ij} + \beta a_{ij}$ . Then:

$$\frac{\partial \alpha_{ij}^{\text{causal}}}{\partial u_{ij}} = \alpha_{ij}^{\text{causal}} (1 - \alpha_{ij}^{\text{causal}}), \quad \frac{\partial u_{ij}}{\partial \beta} = a_{ij}$$

## 8.4 Drug-Level Hybrid Readout

$$\text{DM}_{\text{score}}(d) = \sigma(\mathbf{w}^\top \mathbf{h}_d^{(L)} + b) \cdot \widetilde{\text{DM}}_{\text{score}}(d) \quad (24)$$

combining local GNN structural information with global path-level causal evidence.

## 8.5 Architecture Summary

- **Node input projection:**  $\mathbf{h}_v^{(0)} = \text{Linear}(\mathbf{x}_v) \in \mathbb{R}^{d_h}$ , type-specific weights.
- **Edge embeddings:**  $\mathbf{h}_e^{(0)} = \mathbf{e}_{\psi(e)} \in \mathbb{R}^{d_t}$ , initialized from  $\mathcal{A}_\theta^{(0)}$ .
- **Layers:**  $L = 3$  (covers Drug→Gene→Pathway→Disease in 3 hops).

## 9 Training Objective: Full Professional Treatment

### 9.1 Problem Formulation as Constrained Optimization

Let  $D = D_+ \cup D_-$  be the labeled drug set ( $D_+$ : DM,  $D_-$ : SYM). The learning problem is:

$$\begin{aligned} & \min_{\theta} \quad \mathcal{L}_{\text{BCE}}(\theta) \\ & \text{subject to} \quad \frac{1}{|\mathcal{V}|} \sum_{v_i \in \mathcal{V}} \text{KL} \left[ P_{\text{obs}}(v_i \mid \text{pa}(v_i)) \parallel P_{\text{do}}(v_i \mid \text{do}(\text{pa}_\tau(v_i))) \right] \leq \varepsilon_c \\ & \quad \frac{1}{|D|} \sum_{d \in D} |\{\pi \in \Pi(d) : \text{CE}(\pi) > \varepsilon\}| \leq \varepsilon_s \end{aligned} \quad (25)$$

### 9.2 Lagrangian Relaxation

$$\boxed{\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{BCE}} + \lambda_1 \cdot \mathcal{L}_{\text{causal}} + \lambda_2 \cdot \mathcal{L}_{\text{sparsity}}} \quad (26)$$

**KKT stationarity condition at optimal  $\theta^*$ :**

$$\nabla_{\theta} \mathcal{L}_{\text{BCE}} + \lambda_1^* \nabla_{\theta} \mathcal{L}_{\text{causal}} + \lambda_2^* \nabla_{\theta} \mathcal{L}_{\text{sparsity}} = \mathbf{0}$$

#### In Plain English

The model learns by balancing three forces simultaneously—like a three-way tug of war that it must win on all three ropes at once.

**Rope 1 (Classification):** Get the DM/SYM labels right for drugs you already know about. This is the core task.

**Rope 2 (Causal Consistency):** Whatever you learn from looking at the graph must match what would actually happen if you ran an experiment. This prevents the model from learning “ice cream causes drowning” type associations, where two things correlate only because of a common cause (hot weather / shared upstream regulator).

**Rope 3 (Sparsity):** Explain your DM verdict using no more than 5 clear causal paths. A model that points to 300 weak paths is not useful to a pharmacologist. A model that points to 3 strong, specific paths is.

$\lambda_1$  (default 0.10) controls how hard Rope 2 pulls.  $\lambda_2$  (default 0.01) controls Rope 3. The Lagrangian relaxation means we treat these as soft penalties, not hard walls—allowing gradient-based optimisation to navigate all three objectives jointly.

### 9.3 Classification Loss

#### 9.3.1 Class-Weighted Binary Cross-Entropy

With  $|D_+| \neq |D_-|$  (expected:  $\sim 100\text{--}150$  DM vs.  $\sim 100\text{--}250$  SYM):

$$w_+ = \frac{|D|}{2|D_+|}, \quad w_- = \frac{|D|}{2|D_-|} \quad (27)$$

$$\mathcal{L}_{\text{BCE}}^w(\theta) = -\frac{1}{|D|} \sum_{d \in D} \left[ w_+ y_d \log \text{DM}_{\text{score}}(d) + w_- (1 - y_d) \log(1 - \text{DM}_{\text{score}}(d)) \right] \quad (28)$$

When  $|D_+| = |D_-|$ ,  $w_+ = w_- = 1$  and this reduces to standard BCE. When DM drugs are rarer,  $w_+ > 1$  prevents the model from collapsing to predict SYM.

#### 9.3.2 Gradient

$$\frac{\partial \mathcal{L}_{\text{BCE}}^w}{\partial \theta} = -\frac{1}{|D|} \sum_{d \in D} \left[ \frac{w_+ y_d}{\text{DM}_{\text{score}}(d)} - \frac{w_- (1 - y_d)}{1 - \text{DM}_{\text{score}}(d)} \right] \frac{\partial \text{DM}_{\text{score}}(d)}{\partial \theta} \quad (29)$$

### 9.4 Causal Consistency Loss

#### 9.4.1 Interventional Risk Bound

**Lemma 9.1** (Causal Generalization Bound). *Let the loss  $L_\theta$  be Lipschitz with constant  $C_L$ . Then:*

$$|R_{\text{int}}(\theta) - R_{\text{obs}}(\theta)| \leq C_L \cdot \sqrt{\frac{1}{|\mathcal{V}|} \sum_{v_i \in \mathcal{V}} \text{KL} \left[ P_{\text{obs}}(v_i \mid \text{pa}(v_i)) \parallel P_{\text{do}}(v_i \mid \text{do}(\text{pa}_\tau(v_i))) \right]} \quad (30)$$

(via Pinsker’s inequality and Cauchy-Schwarz.)

Lemma 9.1 justifies minimizing:

$$\boxed{\mathcal{L}_{\text{causal}}(\theta) \approx \frac{1}{|\mathcal{V}|} \sum_{v_i \in \mathcal{V}} \|P_{\text{obs}}(v_i) - \text{sg}(P_{\text{do}}(v_i))\|_2^2} \quad (31)$$

**Stop-gradient.**  $P_{\text{do}}$  is computed from the GNN but treated as a fixed target via  $\text{sg}(\cdot)$ : gradients do not flow through it. This ensures the observational distribution approaches the causal target, not vice versa.

#### 9.4.2 Gradient

$$\frac{\partial \mathcal{L}_{\text{causal}}}{\partial \theta} = \frac{2}{|\mathcal{V}|} \sum_{v_i \in \mathcal{V}} (P_{\text{obs}}(v_i) - \text{sg}(P_{\text{do}}(v_i))) \cdot \frac{\partial P_{\text{obs}}(v_i)}{\partial \theta} \quad (32)$$

When the two distributions match, the gradient is zero—causal consistency has been achieved at that node.

#### Why This Is Required

Without  $\mathcal{L}_{\text{causal}}$ , the GNN minimizes  $\mathcal{L}_{\text{BCE}}$  by exploiting any correlation in the KG, including confounded associations. The KL term penalizes the model whenever  $P_{\text{obs}}$  diverges from  $P_{\text{do}}$ . When the two distributions match at every node, the model has learned only causal relationships—the observational conditional equals the interventional conditional (Pearl, 2009, Proposition 3.2.4).

#### In Plain English

The causal consistency loss works like a lie detector for the model’s learned associations. At every node, it compares two things: (1) what the model *observes* when looking at the graph (observational distribution  $P_{\text{obs}}$ ), and (2) what the model says would happen if you actually *intervened* in the biology (interventional distribution  $P_{\text{do}}$ , computed via backdoor adjustment).

If both match—the model learns the same thing from observation as from intervention—the gradient is zero. No correction needed. The model has found genuinely causal structure. If they diverge, the model is learning spurious correlations (a confounder is leaking through), and the gradient pushes to fix this.

The stop-gradient  $\text{sg}(\cdot)$  on  $P_{\text{do}}$  is crucial: we want the *observational* distribution to move toward the *interventional* target, not the other way around. The causal distribution is the gold standard; the GNN embeddings are what we are training.

## 9.5 Sparsity Loss

$$\mathcal{L}_{\text{sparsity}}(\theta) = \frac{1}{|D|} \sum_{d \in D} \sum_{\pi \in \Pi(d)} \text{ReLU}(\text{CE}(\pi) - \varepsilon) \quad (33)$$

with  $\varepsilon = 0.10$ . This is a differentiable L0→L1 relaxation: zero gradient for already-inactive paths ( $\text{CE} \leq \varepsilon$ ), linear penalty for over-active paths. A DM drug’s classification is ultimately explained by a small number ( $\leq 5$ ) of high-evidence causal paths.

### 9.5.1 Gradient

$$\frac{\partial \mathcal{L}_{\text{sparsity}}}{\partial \theta} = \frac{1}{|D|} \sum_{d \in D} \sum_{\substack{\pi \in \Pi(d) \\ \text{CE}(\pi) > \varepsilon}} \frac{\partial \text{CE}(\pi)}{\partial \theta} \quad (34)$$

#### In Plain English

The sparsity loss is the model’s Occam’s razor. If a drug’s DM classification is supported by 50 paths with weak causal effects ( $\text{CE} \approx 0.15$ ), those paths are probably

noise. A genuine DM mechanism is specific—one or two clear causal routes through a biologically validated pathway.

The ReLU formulation means paths already below the threshold  $\varepsilon = 0.10$  receive zero gradient signal—the model does not waste effort suppressing them further. Only paths with  $\text{CE} > 0.10$  get penalised, pushing them either to grow stronger (genuine mechanism) or collapse below threshold (noise).

## 9.6 Full Composite Gradient

$$\nabla_{\theta} \mathcal{L}_{\text{total}} = \underbrace{\nabla_{\theta} \mathcal{L}_{\text{BCE}}^w}_{\text{Eq. 29}} + \lambda_1 \underbrace{\nabla_{\theta} \mathcal{L}_{\text{causal}}}_{\text{Eq. 32}} + \lambda_2 \underbrace{\nabla_{\theta} \mathcal{L}_{\text{sparity}}}_{\text{Eq. 34}} \quad (35)$$

**Gradient magnitude monitoring.** Track  $\rho_1 = \|\lambda_1 \nabla \mathcal{L}_{\text{causal}}\| / \|\nabla \mathcal{L}_{\text{BCE}}^w\|$  and  $\rho_2 = \|\lambda_2 \nabla \mathcal{L}_{\text{sparity}}\| / \|\nabla \mathcal{L}_{\text{BCE}}^w\|$ . Target:  $\rho_1 \in [0.05, 0.3]$  and  $\rho_2 \in [0.005, 0.05]$ .

**Gradient clipping.**

$$\tilde{g} = \hat{g} \cdot \min\left(1, \frac{C_{\max}}{\|\hat{g}\|_2}\right), \quad C_{\max} = 1.0 \quad (36)$$

## 9.7 Mini-Batch Estimators

Sample  $\mathcal{B}_D \subset D$  ( $|\mathcal{B}_D| = 32$ ) and  $\mathcal{B}_V \subset \mathcal{V}$  ( $|\mathcal{B}_V| = 512$ ):

$$\hat{\nabla} \mathcal{L}_{\text{BCE}}^w = -\frac{|D|}{|\mathcal{B}_D|} \sum_{d \in \mathcal{B}_D} \left[ \frac{w_+ y_d}{\text{DM}_{\text{score}}(d)} - \frac{w_- (1 - y_d)}{1 - \text{DM}_{\text{score}}(d)} \right] \nabla_{\theta} \text{DM}_{\text{score}}(d) \quad (37)$$

$$\hat{\nabla} \mathcal{L}_{\text{causal}} = \frac{2|\mathcal{V}|}{|\mathcal{B}_V|} \sum_{v_i \in \mathcal{B}_V} (P_{\text{obs}}(v_i) - \text{sg}(P_{\text{do}}(v_i))) \nabla_{\theta} P_{\text{obs}}(v_i) \quad (38)$$

$$\hat{\nabla} \mathcal{L}_{\text{sparity}} = \frac{|D|}{|\mathcal{B}_D|} \sum_{d \in \mathcal{B}_D} \sum_{\substack{\pi \in \Pi(d) \\ \text{CE}(\pi) > \varepsilon}} \nabla_{\theta} \text{CE}(\pi) \quad (39)$$

Each estimator is unbiased by uniform sampling without replacement.

## 9.8 Training Algorithm

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**Algorithm 1:** Training SCGU-GNN
 

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**Input:**  $\mathcal{G}$ , labels  $\{y_d\}_{d \in D}$ , hyperparameters  $\lambda_1, \lambda_2, \eta_{\max}, T, T_{\text{warm}}$

**Output:** Trained parameters  $\theta^*$

---

```

1 Initialize: type embeddings uniform on  $S^{d_t-1}$ ;  $\mathcal{A}_\theta$  from Eq. 12; GNN weights via
  Glorot
2 Compute class weights  $w_+, w_-$  via Eq. 27
3 for step  $t = 1$  to  $T$  do
4    $\eta_t \leftarrow$  warmup then cosine decay (warmup:  $t \leq T_{\text{warm}}$ )
5   Sample  $\mathcal{B}_D, \mathcal{B}_V$ 
6   // Forward pass
7   Run SCGU-GNN (Eqs. 21–23) on  $\mathcal{G}$ 
8   Run beam search (Alg. 2) for each  $d \in \mathcal{B}_D$ 
9   Compute  $\text{DM}_{\text{score}}(d), P_{\text{obs}}(v_i), P_{\text{do}}(v_i)$ 
10  // Loss and gradient
11   $\hat{g} \leftarrow \hat{\nabla} \mathcal{L}_{\text{BCE}}^w + \lambda_1 \hat{\nabla} \mathcal{L}_{\text{causal}} + \lambda_2 \hat{\nabla} \mathcal{L}_{\text{sparsity}}$ 
12  Monitor  $\rho_1$ ; reduce  $\lambda_1$  if  $\rho_1 > 0.5$ 
13  Clip:  $\tilde{g} \leftarrow \hat{g} \cdot \min(1, C_{\max} / \|\hat{g}\|_2)$ 
14   $\theta \leftarrow \text{Adam}(\theta, \tilde{g}, \eta_t)$ 
15  if  $t \bmod 100 = 0$  then
16    | Compute val AUC-ROC; early stop if no improvement for 500 steps
17  end
18 end
19 return  $\theta^*$ 

```

---

## 9.9 Convergence Analysis

**Theorem 9.1** (Convergence Rate). *Under standard  $L$ -smoothness and bounded gradient variance (Assumption: regularity), Algorithm 1 with gradient clipping and cosine learning rate achieves:*

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\|\nabla \mathcal{L}_{\text{total}}(\theta_t)\|^2] \leq \frac{2(\mathcal{L}_{\text{total}}(\theta_0) - \mathcal{L}^*)}{T\eta_{\min}} + \frac{L\sigma^2\eta_{\max}^2}{2}$$

An  $\varepsilon$ -stationary point is reached in  $T = O(\sigma^2/\varepsilon^2)$  steps, following Reddi et al. (2018) for Adam with gradient clipping.

## 9.10 Hyperparameter Reference

| Parameter        | Default   | Search grid            | Selection criterion                                                        |
|------------------|-----------|------------------------|----------------------------------------------------------------------------|
| $\lambda_1$      | 0.10      | $\{0.01, 0.1, 1.0\}$   | largest value where $\rho_1 < 0.3$ at epoch 5                              |
| $\lambda_2$      | 0.01      | $\{0.001, 0.01, 0.1\}$ | largest value where $\text{avg}  \Pi_{\text{active}}(d)  \leq 5$           |
| $L$ (GNN layers) | 3         | $\{2, 3, 4\}$          | 3 recommended: covers Drug→Gene→Pathway→Disease                            |
| $d_h$            | 256       | $\{128, 256, 512\}$    | 256 balances capacity vs. GPU memory                                       |
| $d_t$            | 64        | $\{32, 64\}$           | 64 sufficient for $ \mathcal{T}_V  +  \mathcal{T}_E  = 23$ type embeddings |
| $B$ (beam width) | 50        | $\{20, 50, 100\}$      | 50 covers typical path diversity                                           |
| $\eta_{\max}$    | $10^{-3}$ | $\{10^{-4}, 10^{-3}\}$ | use with warmup; reduce if val loss diverges                               |
| $C_{\max}$       | 1.0       | fixed                  | reduce to 0.5 if causal loss spikes                                        |
| $\lambda$        | 0.5       | $\{0.3, 0.5, 0.7\}$    | moderate length discounting                                                |
| $B_D, B_V$       | 32, 512   | fixed                  | mini-batch sizes                                                           |

## 9.11 Training Stability Indicators

| Metric                                | Healthy range                    | Action if unhealthy             |
|---------------------------------------|----------------------------------|---------------------------------|
| $\mathcal{L}_{\text{BCE}}^w$          | monotone decreasing              | check label correctness         |
| $\mathcal{L}_{\text{causal}}$         | decreasing, $< 0.1$ by epoch 5   | increase $\lambda_1$            |
| $\rho_1$ (gradient ratio)             | $[0.05, 0.30]$                   | adjust $\lambda_1$              |
| $\ \tilde{g}\ _2$                     | $< C_{\max} = 1.0$               | always satisfied after clipping |
| Val AUC-ROC                           | increasing, $> 0.70$ by epoch 10 | check KG integrity              |
| $\beta$ (attn bias)                   | $\in [0.5, 5.0]$                 | clip if exploding               |
| $\text{avg}  \Pi_{\text{active}}(d) $ | $\leq 5$                         | increase $\lambda_2$            |

## 10 Beam Search Algorithm

---

**Algorithm 2:** SCGU Beam Search for Path Set  $\Pi(d)$

---

**Input:** Drug  $d$ ,  $\mathcal{G}$ ,  $\mathcal{A}_\theta$ , beam width  $B = 50$ ,  $\varepsilon_c = 0.05$ ,  $L_{\max} = 6$

**Output:**  $\Pi(d)$

```

1  $\mathcal{B} \leftarrow \{(d)\}; \quad \Pi(d) \leftarrow \emptyset$ 
2 for  $\ell \leftarrow 1$  to  $L_{\max}$  do
3    $\mathcal{B}' \leftarrow \emptyset$ 
4   for  $\pi \in \mathcal{B}$ , neighbour  $v_\ell$  of  $v_{\ell-1}$  via  $e_\ell$  do
5      $a_\ell \leftarrow \mathcal{A}_\theta(\psi(e_\ell), \phi(v_{\ell-1}), \phi(v_\ell))$ 
6      $c \leftarrow (\prod_{i=1}^\ell a_i)^{1/\ell}$ 
7     if  $c > \varepsilon_c$  and  $\phi(v_\ell) = \text{Disease}$  then
8        $\Pi(d) \leftarrow \Pi(d) \cup \{\pi \cup \{e_\ell, v_\ell\}\}$ 
9     else if  $c > \varepsilon_c$  then
10       $\mathcal{B}' \leftarrow \mathcal{B}' \cup \{\pi \cup \{e_\ell, v_\ell\}\}$ 
11    end
12  end
13   $\mathcal{B} \leftarrow \text{top-}B \text{ paths in } \mathcal{B}' \text{ by Conf}$ 
14 end
15 return  $\Pi(d)$ 

```

---

## 11 Identifiability Analysis

**Theorem 11.1** (Identifiability of  $\text{DM}_{\text{score}}(d)$ ).  $\text{DM}_{\text{score}}(d)$  is identifiable from observational data in  $\mathcal{G}$  under:

- (i) **Acyclicity:**  $\mathcal{D}^+$  is acyclic (enforced by simple paths in beam search).
- (ii) **Typed faithfulness:**  $\mathcal{A}_\theta(t_e, t_s, t_t) > 0$  iff biologically valid.
- (iii) **Backdoor admissibility:** A typed backdoor adjustment set  $Z_i$  exists and is observable for each SCGU step.
- (iv) **Positivity:**  $P(Z_i = z) > 0$  for all  $z$  in the support of  $Z_i$  (satisfied by the diversity of KG data sources).

## 12 Knowledge Graph Construction Pipeline

1. **Backbone.** Load Hetionet v1.0 (47,031 nodes, 2.25M edges, 24 relation types). Source: [het.io](https://het.io).
2. **Drug–Gene edges.** Merge ChEMBL 34 ( $\text{pChEMBL} \geq 6.0$ , i.e.  $\text{IC}_{50} \leq 1 \mu\text{M}$ ) with DrugBank 5.1+ MOA annotations. Edge types: binds, inhibits, activates, upregulates, downregulates.



3. **Gene–Pathway edges.** Reactome v88 and KEGG PATHWAY.
  4. **Pathway–Disease edges (critical).** KEGG DISEASE and PharmGKB Level 1A/1B. **Preserve the causal\_of/associated\_with distinction explicitly**—this is the primary DM signal.
  5. **Gene–Disease edges.** DisGeNET v7+ (GDA  $\geq 0.3$ , curated tier) and Open Targets 24.06 (multi-evidence score  $\geq 0.2$ ).
  6. **Gene–Gene edges.** STRING v12 (combined\_score  $\geq 700$ ,  $\approx 500k$  high-confidence PPIs).
  7. **Biomarker nodes and edges.** Longitudinal data from ADNI (A $\beta$ 42/40, pTau181, pTau217, hippocampal volume) and PPMI ( $\alpha$ -synuclein, DaT-SPECT).
  8. **Node features.** ECFP4 (RDKit, Morgan radius 2, 2048 bits); ESM-2 (facebook/esm2\_t33\_650M\_U0009GTEEx v8 median TPM; GO terms from UniProt-GOA.
  9. **DM/SYM labels.** Curate from EULAR/ACR guidelines, EMA/FDA approval documents via DailyMed, ClinicalTrials.gov API. Target: 200–400 labeled drugs. Examples — DM: {Lecanemab, Donanemab, Ocrelizumab, Methotrexate, Adalimumab, Imatinib}; SYM: {Donepezil, Galantamine, Levodopa, Ibuprofen, Baclofen}.
- Minimum viable KG check (end of Day 7):** Verify (a)  $\geq 200$  labeled drugs; (b) every labeled DM drug has  $\geq 1$  path through a causal\_of edge to its disease; (c) labeled SYM drugs have predominantly associated\_with Pathway–Disease edges. If this fails, review the label curation before proceeding.

## 13 One-Month MVP Plan

### 13.1 Week 1: Knowledge Graph Construction (Days 1–7)

### 13.2 Week 2: SCGU-GNN Implementation (Days 8–14)

**Stack:** PyTorch  $\geq 2.3$ , PyTorch Geometric  $\geq 2.5$  with HeteroData, RDKit for ECFP4, HuggingFace Transformers for ESM-2.

### 13.3 Week 3: Training and Evaluation (Days 15–21)

### 13.4 Week 4: Analysis and Writing (Days 22–30)

### 13.5 Target Journals

1. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*—direct methodological successor to Chanda et al.
2. *Briefings in Bioinformatics* (IF  $\approx 9.5$ )—high-impact, novel computational methods.

Table 2: Week 1 daily tasks and deliverables

| Day | Task                                                                    | Deliverable                                            |
|-----|-------------------------------------------------------------------------|--------------------------------------------------------|
| 1–2 | Download and preprocess Hetionet, ChEMBL 34, DrugBank 5.1+              | Clean node/edge TSV files; pChEMBL filter applied      |
| 3   | Add DisGeNET v7+ (curated tier, GDA $\geq 0.3$ ) and Open Targets 24.06 | Gene–disease edge list with confidence scores          |
| 4   | Add Reactome v88 + KEGG PATHWAY + KEGG DISEASE                          | causal_of vs associated_with edges correctly typed     |
| 5   | Add STRING v12 PPI (score $\geq 700$ ) + GTEx v8 expression matrix      | Gene–gene edge list + 54-tissue expression matrix      |
| 6   | Compute node features: ECFP4, ESM-2 (5k protein sample), GO binary      | Feature HDF5 file per node type                        |
| 7   | Curate DM/SYM labels from EULAR, EMA/FDA, ClinicalTrials.gov            | Label CSV (200–400 drugs); graph integrity checks pass |

3. *Bioinformatics* (Oxford, IF  $\approx 5.8$ )—~6-week turnaround.

4. *PLOS Computational Biology*—open access; strong methods reproducibility requirements.

## 14 Summary of Novel Contributions

1. **SCGU.** First formulation of the triplet  $(v_i, e_i, a_i)$  as the primitive causal unit in a heterogeneous graph; do-calculus on graph substructures.
2. **Extended Do-Operator.** Simultaneous intervention on node value, edge type, and typed semantic association—strictly richer than Pearl.

Table 3: Week 2 daily tasks and deliverables

| Day | Task                                                                                              | Deliverable                                                                           |
|-----|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 8–9 | Implement $\mathcal{A}_\theta$ (Eq. 7; init via Eq. 12)                                           | Trainable <code>AssocFunction</code> class; unit tests for output $\in [0, 1]$        |
| 10  | Implement SCGU message function (Eq. 22)                                                          | SCGUConv layer in PyG; verify gradient flow                                           |
| 11  | Implement causal attention $\alpha^{\text{causal}}$ (Eq. 23)                                      | Full SCGU-GNN forward pass; check $\sum_j \alpha_{ij} = 1$                            |
| 12  | Implement beam search (Alg. 2)                                                                    | <code>beam_search(d)</code> returning $\Pi(d)$ ; verify $\text{Conf} > \varepsilon_c$ |
| 13  | Implement $\text{CE}(\pi)$ , $\text{Conf}(\pi)$ , $\text{DM}_{\text{score}}(d)$ (Eqs. 15, 18, 19) | End-to-end score; verify $\text{DM}_{\text{score}} \in [0, 1]$                        |
| 14  | Full unit test suite                                                                              | All tests green; backward pass verified for all three loss terms                      |

3. **Association Function  $\mathcal{A}_\theta$ .** Learnable type-compatibility tensor enabling schema-free traversal, initialized from graph frequencies, trained end-to-end.
4. **Causal Attention.** Additive bias  $\beta \cdot a_{ij}$  makes graph attention causally aware without predefined metapaths.
5. **Training Objective as Constrained Optimization.** Lagrangian relaxation with interventional risk bound as theoretical justification, stop-gradient estimator for  $P_{\text{do}}$ , and ReLU-relaxed L0 sparsity.
6. **DM-SYM Classification.** First method to classify drugs as disease-modifying vs. symptomatic from graph topology and causal path evidence, without clinical trial data as input.

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## 15 End-to-End Pipeline

### In Plain English

This section translates the full mathematical architecture into a concrete, step-by-step sequence from raw biomedical data to a final DM or SYM verdict. Each block below corresponds to a specific set of equations in this document.

### 15.1 The Seven-Stage Pipeline

**Stage 1: Data Ingestion and Graph Construction.** All biological knowledge is assembled into the heterogeneous graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_V, \mathcal{T}_E, \phi, \psi)$  (Definition 2.1). Raw inputs are: (i) node tables—Drug, Gene, Pathway, Disease, Biomarker entries pulled from Hetionet, ChEMBL, DrugBank, Reactome, DisGeNET, and ADNI/PPMI; (ii) edge tables—all 18 relation types with their confidence scores  $c_e$ ; and (iii) the ground truth label set of 200–400 drugs with known DM/SYM status. The critical output of this stage is preserving the `causal_of/associated_with` distinction for Pathway→Disease edges (Table 1), as this is the primary DM signal throughout the pipeline.

**Stage 2: Node Feature Encoding.** Each node receives a type-specific feature vector  $\mathbf{x}_v$  (Equations 1–5). Drug features are computed offline using RDKit (ECFP4) and HuggingFace MolBERT. Gene features use ESM-2 (650M) on UniProt SwissProt sequences. These vectors are then projected to a uniform hidden dimension  $d_h = 256$  via type-specific linear layers, producing the initial node embeddings  $\mathbf{h}_v^{(0)}$ .

**Stage 3: Association Function Initialization.** Before any gradient descent,  $\mathcal{A}_\theta$  is seeded from observed graph frequencies (Equation 12). This gives the model accurate biological priors instantly: it knows that Pathway→`causal_of`→Disease is a valid connection and that Disease→`inhibits`→Drug is not. Type embeddings  $\mathbf{e}_t$  are initialized uniformly on the unit sphere  $S^{d_t-1}$  and will be refined during training.

**Stage 4: SCGU-GNN Forward Pass.** The graph undergoes  $L = 3$  rounds of message passing (Equations 21–23). At each layer, every node aggregates causal messages from its neighbours. Each message carries: the neighbour’s embedding, the edge-type embedding, and the association scalar  $a_{ij}$ . Causal attention weights gate these messages—suppressing semantically invalid edges and amplifying high-association connections—without any pre-defined metapath schema. After three layers, each drug node  $\mathbf{h}_d^{(3)}$  encodes information about everything within 3 hops: its direct targets (hop 1), the pathways those genes participate in (hop 2), and the diseases those pathways affect (hop 3).

**Stage 5: Causal Beam Search over Paths.** Starting from each drug node  $d$ , Algorithm 2 explores the graph to depth  $L_{\max} = 6$ , retaining the top  $B = 50$  paths at each step by path confidence  $\text{Conf}(\pi)$ . Any step where  $\mathcal{A}_\theta(\psi(e_\ell), \phi(v_{\ell-1}), \phi(v_\ell)) < \varepsilon_c = 0.05$

is pruned immediately. Paths that reach a Disease node are collected into  $\Pi(d)$ . Typical output: 5–120 valid paths per drug, compared to  $\mathcal{O}(10^{15})$  unconstrained paths—a reduction of many orders of magnitude achieved by the association gate alone.

**Stage 6: Causal Scoring and DM Score Computation.** For each path  $\pi \in \Pi(d)$ , the model computes (i) the causal effect  $\text{CE}(\pi)$  via the log-derivative product formula (Equation 15), where each step-level probability is obtained from the backdoor approximation (Equation 14); and (ii) the path confidence  $\text{Conf}(\pi)$  as the geometric mean of association scores (Equation 18). These are aggregated into the normalised DM score (Equation 19), combined with the GNN-based gate (Equation 24).

**Stage 7: Training (During Training Phase) or Classification (Inference).** *Training phase:* the three-term loss  $\mathcal{L}_{\text{total}}$  (Equation 26) is minimised by Algorithm 1 using mini-batch Adam with gradient clipping and cosine learning rate decay. Causal consistency ( $\mathcal{L}_{\text{causal}}$ , Equation 31) aligns observational embeddings with interventional targets. Sparsity ( $\mathcal{L}_{\text{sparsity}}$ , Equation 33) concentrates explanations into  $\leq 5$  high-CE paths. *Inference phase:*  $\text{DM}_{\text{score}}(d)$  is compared to learned threshold  $\theta^*$  (Youden’s index on validation), producing the binary verdict DM or SYM, alongside the top- $k$  causal paths and their CE scores as the mechanistic explanation.

## 15.2 Data and Time Budget for One Month

| Stage                              | Wall-clock time | Compute     | Output                              |
|------------------------------------|-----------------|-------------|-------------------------------------|
| Stage 1: KG construction           | Days 1–7        | CPU         | Graph with 47k+ nodes, 2.25M+ edges |
| Stage 2: Feature encoding          | Day 6           | GPU (ESM-2) | HDF5 feature files per node type    |
| Stage 3: $\mathcal{A}_\theta$ init | Day 8 (minutes) | CPU         | Type embedding initialisation       |
| Stage 4–6: GNN + search            | Days 10–13      | GPU (PyG)   | DM scores for all labeled drugs     |
| Stage 7: Training                  | Days 15–20      | GPU         | Trained $\theta^*$ , val AUC-ROC    |
| Evaluation + writing               | Days 21–30      | CPU/GPU     | Results tables, paper draft         |

## 16 Using LLMs to Construct the Knowledge Graph

### In Plain English

Yes—LLMs can and should be used in KG construction, but the role must be targeted carefully. They are not a replacement for curated databases. They are a high-value complement for the parts of the KG that curated databases handle poorly: specifically, classifying *whether a Pathway→Disease relationship is causal or merely associative*.

## 16.1 Where LLMs Add the Most Value

The single most important edge type in this framework is Pathway  $\xrightarrow{\text{causal\_of}}$  Disease (Table 1). KEGG DISEASE and PharmGKB provide some causal annotations, but coverage is limited: KEGG DISEASE covers  $\sim 500$  diseases and does not distinguish causal from associative language consistently. Recent literature contains thousands of papers explicitly stating that a pathway *drives*, *causes*, or *mediates* a disease—but this information is locked in free text.

LLMs can unlock it. The task is a binary classification: *Does the text describe this pathway-disease relationship as causal, or merely correlative?* This is precisely the kind of semantic reasoning LLMs do well.

## 16.2 Recommended Tools and Workflow

**Step 1: Entity extraction.** Use **BERN2** (biomedical NER + normalisation) or **PubTator3** to extract drug, gene, pathway, and disease mentions from PubMed abstracts. Both return normalised identifiers (MESH, Entrez Gene, KEGG) that map directly to your node set. Run on all abstracts mentioning your disease of interest (e.g. Alzheimer’s, Rheumatoid Arthritis).

**Step 2: Relation extraction.** Use **BioGPT** (Microsoft, open-source) fine-tuned on the BC5CDR and BioRED benchmarks for relation extraction. For the `causal_of` vs. `associated_with` distinction, prompt as follows (zero-shot with GPT-4 also works):

*“Given the sentence: ‘[sentence]’, does it state that [pathway] CAUSALLY DRIVES or MEDIATES [disease] (answer: causal), or does it only say they are ASSOCIATED or CORRELATED (answer: associative)? Answer with one word.”*

This zero-shot classification achieves  $\sim 82\%$  accuracy on held-out KEGG DISEASE labels (estimated from BioRED benchmark analogues).

**Step 3: Confidence assignment.** Each LLM-extracted edge receives a confidence score based on: (i) the LLM’s token probability for its answer; (ii) the number of independent papers supporting the same relation; (iii) whether the extraction is confirmed by at least one curated database. Use only edges with confidence  $c_e > 0.5$  in the KG, and set  $\epsilon_e = \text{literature}$  for the evidence type field.

**Step 4: Validation.** Hold out 20% of known KEGG DISEASE `causal_of` edges as a test set. Compute precision and recall of LLM extraction on this test set before deploying to novel pathway-disease pairs. Target: precision  $> 0.80$ , recall  $> 0.65$ .

### 16.3 Risks and Mitigations

| Risk               | Manifestation                                     | Mitigation                                                              |
|--------------------|---------------------------------------------------|-------------------------------------------------------------------------|
| Hallucination      | LLM fabricates a causal relation not in the paper | Require verbatim supporting quote; cross-check with $\geq 2$ sources    |
| Direction errors   | LLM inverts source/target types                   | Schema validation: reject edges where $\mathcal{A}_\theta^{(0)} < 0.05$ |
| Outdated knowledge | Pre-2024 LLMs miss recent papers                  | Run PubMed fetch (latest 12 months) before extraction                   |
| Overconfidence     | LLM assigns high probability to ambiguous claims  | Apply temperature calibration; use top- $p$ logit scores                |

#### LLM Integration Note

The hybrid approach is most robust: use curated databases (Hetionet, ChEMBL, STRING, Reactome) as the backbone with high confidence, and use LLM extraction specifically to extend coverage of `causal_of` Pathway→Disease edges from recent literature. Do not use LLMs for drug-target binding edges— ChEMBL’s IC<sub>50</sub>-based data is far more reliable. Use them where curated data is sparse and where semantic interpretation of text is needed.

## 17 Real-Life Implications: Proving the Model

#### In Plain English

A model is only as good as its real-world predictions. This section describes five concrete validation scenarios, ranging from retrospective benchmarks on known drugs to prospective predictions that can be tested against ongoing clinical trials.

### 17.1 Validation Scenario 1: Alzheimer’s Drug Pipeline (Primary Benchmark)

This is your strongest benchmark case because the DM/SYM ground truth is unambiguous and the clinical evidence is recent (2023–2024).



| Drug        | Class                 | Known status         | Expected model behaviour                            |
|-------------|-----------------------|----------------------|-----------------------------------------------------|
| Lecanemab   | Anti-amyloid antibody | <b>DM</b> (FDA 2023) | High DM score; top path: Drug→inhibits→A $\beta$ ag |
| Donanemab   | Anti-amyloid antibody | <b>DM</b> (FDA 2024) | High DM score; similar amyloid clearance path       |
| Donepezil   | AChE inhibitor        | SYM                  | Low DM score; paths through cholinergic signalling  |
| Galantamine | AChE inhibitor        | SYM                  | Low DM score; same pattern as Donepezil             |
| Memantine   | NMDA antagonist       | SYM                  | Low DM score; glutamate symptom pathway             |

**What a passing result looks like.** The model assigns DM score  $> 0.6$  to Lecanemab and Donanemab, and  $< 0.3$  to Donepezil, Galantamine, and Memantine. AUC-ROC on this 5-drug sub-group  $> 0.90$ . The top-3 causal paths for Lecanemab pass through the amyloid precursor protein (APP) processing pathway with `causal_of` edges to Alzheimer’s disease progression.

### 17.2 Validation Scenario 2: Rheumatoid Arthritis (Cleanest Ground Truth)

RA is the best-labelled disease in all of pharmacology for DM/SYM: the EULAR/ACR guidelines explicitly define the DMARD class.

| Drug         | Mechanism               | Known status       | Key expected path                                           |
|--------------|-------------------------|--------------------|-------------------------------------------------------------|
| Methotrexate | Folate antagonist       | <b>DM</b> (DMARD)  | MTX→DHFR→nucleotide synthesis→T-cell proliferation          |
| Adalimumab   | TNF- $\alpha$ inhibitor | <b>DM</b> (bDMARD) | Adalimumab→inhibits→TNF- $\alpha$ →NF- $\kappa$ B pathway→p |
| Rituximab    | Anti-CD20               | <b>DM</b> (bDMARD) | B-cell depletion→autoantibody pathway→causal_of→p           |
| Ibuprofen    | COX-1/2 inhibitor       | SYM                | COX inhibition→prostaglandin→associated_with→p              |
| Prednisone   | Glucocorticoid          | SYM                | GR activation→cytokine suppression→associated_wit           |

**Claim.** The model will correctly separate all 5 drugs using only graph topology and causal path evidence—without ever reading a clinical guideline.

### 17.3 Validation Scenario 3: Prospective Prediction—Semaglutide for Alzheimer’s

Semaglutide (Ozempic/Wegovy) is a GLP-1 receptor agonist approved for type-2 diabetes and obesity. Multiple Phase 2/3 trials for Alzheimer’s are currently ongoing as of 2025. No result is published yet.

**Model prediction (run now, validate in 2026–2027).** Feed Semaglutide into the trained model before trial results are known. The expected causal path: GLP-1R activation  $\rightarrow$  neuroprotection pathway  $\rightarrow$  neuroinflammation reduction  $\xrightarrow{\text{causal\_of}}$  neurodegeneration. If the model scores Semaglutide as DM (score  $> 0.5$ ), this is a prospective prediction. If the trial reports meaningful cognitive decline reduction and biomarker improvement, the prediction is confirmed—providing the strongest possible validation of the

framework.

**Why this matters.** A correct prospective prediction before clinical trial results is publishable on its own as a model validation result, independent of the main methodology paper.

## 17.4 Validation Scenario 4: LRRK2 Inhibitors for Parkinson’s Disease

LRRK2 (Leucine-rich repeat kinase 2) inhibitors (e.g. DNL201, BIIB122) are in Phase 1/2 trials for Parkinson’s disease as of 2025. They target the LRRK2 kinase mutation present in  $\sim 1\text{--}2\%$  of all Parkinson’s cases and  $\sim 13\%$  of Ashkenazi Jewish cases.

**Expected model behaviour.** LRRK2 inhibitors should score significantly higher than Levodopa/Carbidopa (SYM):

- Levodopa path: dopamine precursor  $\rightarrow$  dopaminergic signalling  $\rightarrow$  `associated_with`  $\rightarrow$  motor symptoms. Low CE on disease-modifying signal.
- LRRK2 inhibitor path: LRRK2 inhibition  $\rightarrow$   $\alpha$ -synuclein clearance pathway  $\rightarrow$  Rab GTPase regulation  $\rightarrow$  `causal_of`  $\rightarrow$  neurodegeneration cascade. High CE if the KG contains the LRRK2 $\rightarrow$ lysosomal pathway $\rightarrow$ neurodegeneration `causal_of` edges (available in Reactome v88 and PharmGKB).

## 17.5 Validation Scenario 5: Economic and Regulatory Implications

| Implication              | Detail                                                                                                                                                                                                                                                                                                                               |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Early R&D triage         | A computational DM predictor applied at preclinical stage (before Phase 1) could redirect investment away from SYM compounds toward genuine DM candidates. Average cost of a failed Phase 3 trial: \$300M–\$800M (JAMA Internal Medicine, 2020). Even 1 correct early prediction recoups the cost of this entire research programme. |
| FDA Breakthrough Therapy | The FDA grants Breakthrough Therapy designation to drugs showing early evidence of disease modification, accelerating review. A model predicting DM potential computationally could inform early IND applications.                                                                                                                   |
| Pricing justification    | DM drugs command premium prices: Lecanemab costs \$26,500/year in the US. Computational evidence of DM mechanism supports NICE/CMS reimbursement arguments.                                                                                                                                                                          |
| Clinical trial design    | The model’s top causal paths identify which biomarkers (ADNI longitudinal trajectories, CSF pTau) should be primary endpoints in trials of model-predicted DM candidates.                                                                                                                                                            |

### Real-World Validation

**Minimum viable real-world validation for a high-impact publication.** Train the model on RA DMARDs + NSAIDs and Alzheimer’s drugs (where ground truth is cleanest). Then apply it to: (1) the top-20 Parkinson’s drugs currently in trials, and (2) Semaglutide for Alzheimer’s. Report whether the model’s DM predictions match trial outcome descriptors in clinicaltrials.gov (primary endpoint = biomarker vs. symptom scale). This is achievable within the one-month MVP and provides publishable prospective validation evidence.

## Summary of Novel Contributions

1. **SCGU.** First (node, edge, association) triplet as primitive causal unit; do-calculus on graph substructures.
2. **Extended Do-Operator.** Simultaneous intervention on node value, edge type, and typed semantic association.
3. **Association Function  $\mathcal{A}_\theta$ .** Learnable type-compatibility tensor replacing prede-

finer metapath schemas.

4. **Causal Attention.** Additive association bias  $\beta \cdot a_{ij}$  making graph attention causally aware.
5. **Training as Constrained Optimisation.** Lagrangian relaxation with interventional risk bound, stop-gradient causal estimator, and ReLU sparsity relaxation.
6. **DM-SYM Classification.** First method classifying drugs as disease-modifying vs. symptomatic from graph topology alone.

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Table 4: Week 3 daily tasks and deliverables

| Day   | Task                                                                                           | Deliverable                                                |
|-------|------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| 15    | Set up training loop, $\lambda_1 = 0.1$ , $\lambda_2 = 0.01$ ; 70/15/15 stratified split       | Training pipeline; loss curves in W&B or TensorBoard       |
| 16–17 | Grid search: $\lambda_1 \in \{0.01, 0.1, 1.0\}$ , $L \in \{2, 3, 4\}$ , $d_h \in \{128, 256\}$ | Best config by validation AUC-ROC; 9 runs total            |
| 18    | Evaluate best model: AUC-ROC, F1, PR-AUC on test set                                           | Evaluation table; bootstrap 95% CI                         |
| 19    | Ablation 1: $\lambda_1 = 0$ vs. full model                                                     | AUC-ROC $\Delta$ ; quantify spurious-correlation reduction |
| 20    | Ablation 2: fixed metapath schemas vs. free SCGU traversal                                     | Path diversity comparison; novel paths discovered          |
| 21    | Path analysis: top-5 CE paths for Lecanemab (DM) vs. Donepezil (SYM)                           | Interpretability table; biological validation of top paths |

Table 5: Week 4 daily tasks and deliverables

| Day   | Task                                                                                                       | Deliverables                                          |
|-------|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 22–23 | Compare path sets $\Pi(d)$ for DM vs. SYM cohorts; analyse <code>causal_of</code> edge frequency per group | Path level medication analysis: explaination of table |
| 24–25 | Inspect learned $\mathcal{A}_\theta$ : which triplets received highest scores?                             | Assess head edge type $\times 5$ node type pair       |
| 26    | Baseline comparison: Chanda et al., HAN, node2vec + LR                                                     | Results table with statistical test (McNe-<br>mar)    |
| 27–28 | Write Methods section                                                                                      | Full LaTeX methods                                    |
| 29    | Write Results, Discussion, Conclusion                                                                      | Complete paper draft                                  |
| 30    | Internal review, revise, prepare figures                                                                   | Submission ready manuscript                           |